



Safety glass

Brittleness

Some materials, such as glass, are very brittle and will break when pushed out of shape. Safety glass is designed to crack rather than break.

Hardness

A scientist called Friedrich Mohs created a scale of ten minerals to compare how hard they are. Many materials are graded on this scale.



Hands on

Collect some different pebbles and put them in order of hardness. A pebble is harder than another if it scratches it. This is how Mohs worked out his scale.

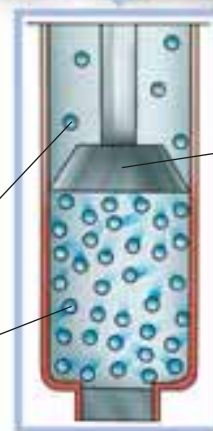
A smooth flow

Some liquids flow more easily than others. It depends on their "stickiness", or viscosity. Hot lava from a volcano flows slowly because it is sticky.

Compressibility

Gases can be squashed, or compressed, by squeezing more into the same space. This is what happens when you pump up a tyre.

Gas can be compressed because its particles are far apart. A bicycle pump pushes the particles closer together.



Foot pump

Gas particles

Diamond is the hardest mineral.

9 Corundum

10 Diamond